SMART SUSTAINABLE WASTE SORTING SYSTEM FOR BEVERAGE CONTAINERS: A TECHNICAL ANALYSIS OF AI-DRIVEN INDUSTRIAL AUTOMATION IN SOUTH AFRICA

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Project Topic: Smart and Sustainable Waste Sorting System for Beverage Containers

Supervisor: Prof. Turgay Celik

Abstract: This technical report presents a detailed design and analysis of a smart waste sorting system designed to address South Africa's waste management challenges through automation technologies. The system integrates multi-modal sensing technologies including computer vision and near-infrared (NIR) spectroscopy with artificial intelligence classification algorithms to achieve sorting performance of 98% accuracy at throughput rates of 180 items per minute. The proposed solution addresses engineering challenges through integration of YOLOv11 object detection, convolutional neural network-based spectral analysis, robotic sorting mechanisms, and industrial IoT infrastructure for active federated learning. Economic analysis demonstrates financial viability with projected payback periods of 1-5 years, while environmental impact assessment indicates substantial contributions to circular economy objectives through enhanced material recovery rates and reduced contamination levels.

Keywords: Computer Vision, Hyperspectral Imaging, Automation, Cybersecurity.

1. INTRODUCTION

Global waste generation continues escalating with environmental concerns as municipal waste production reaches 2.01 billion tons annually and is projected to increase by 70% to 3.4 billion tons by 2050 [1]. Inefficient waste sorting operations contribute to environmental degradation through contaminated recycling streams, reduced material recovery rates, and increased landfill disposal. Traditional manual sorting systems achieve variable accuracy rates of 60-75% while exposing workers to health hazards including respiratory issues from dust exposure, repetitive strain injuries, and contamination risks [2, 3]. These operational limitations necessitate technological solutions addressing social and environmental sustainability as well as its financial viability. The monetary opportunity for automated waste sorting in South Africa is compelling, with the broader recycling economy worth R15 billion annually [4]. The market demand for recycled items is constrained by inadequate sorting capabilities, wherein, the informal recycling sector supports waste reclaimers, however, these individuals operate with limited technological support and significant health and safety risks

The smart waste sorting system analysed in this report seeks to mitigate these issues by implementing multi-modal sensing technologies integrating computer vision and Near-Infrared (NIR) spectroscopy with artificial intelligence (AI) classification algorithms. The system achieves performance 98% average accuracy at throughput rates of 180 items per minute through integration of You Only Look Once (YOLO) object detection, Convolutional Neural Network (CNN) spectral analysis, robotic arm sorting mechanisms and cloud infrastructure to enable active federated learning to accommodate system input dynamics. Economic analysis demonstrates financial viability with projected payback periods of 1- 5 years, while environmental impact assessment indicates substantial contributions to reduced landfill waste and social impact of protecting individuals from the harmful byproducts of waste material. This report presents a detailed investigation into the projects context in section 2, a high-level technical analysis of system architecture in section 3, a detailed system breakdown in section 4, performance evaluation in section 5, financial analysis in section 6 and implementation considerations for South African deployment conditions in section 7 and 8. The analysis addresses engineering requirements including multi-modal sensor integration, real-time processing, communication protocols, and economic viability for successful technology implementation.

2. LITERATURE REVIEW

2.1 Waste Sorting Systems in South Africa

South Africa's waste management infrastructure currently relies predominantly on manual sorting processes, resulting in low recycling rates of approximately 10% compared to global best practices exceeding 50%. The country generates 122 million tonnes of waste annually, with significant portions of valuable recyclable materials ending up in landfills due to inadequate sorting capabilities [3, 5, 6]. Current recycling facilities face challenges including high labour costs, inconsistent sorting quality, and limited processing capacity, creating opportunities for automated solutions to improve both efficiency and material recovery rates. The value proposition for automated sorting is compelling with aluminium cans R9.91 per kilogram, steel containers R6.50-R7.00 per kilogram, and high-quality PET plastics achieving 50-100% of virgin resin prices in export markets [7]. These market values, combined with South Africa's substantial waste volumes, justify the investment in automated sorting technologies that can achieve higher purity rates improved material recovery, protect people and ecosystems from landfill contamination of waterways and gas emissions from waste incineration.

2.2 Waste Object Identification

AI object and material identification for waste sorting presents technical challenges due to the diverse shapes, sizes, colours and conditions of beverage containers found in waste streams. Computer vision systems must handle variations in lighting conditions, object orientations, contamination levels and packaging label differences while maintaining high accuracy rates. Recent advances in deep learning, particularly YOLO architectures, have demonstrated capabilities in real-time object detection applications, with YOLOv11 achieving 97% accuracy and 15ms inference times suitable for industrial sorting applications [8, 9].

Different sensor technologies provide complementary material identification capabilities. Computer vision is capable of performing shape and colour feature recognition but struggles with material composition identification, particularly for transparent or dark-coloured containers. Near Infrared (NIR) spectroscopy (900-1700nm wavelength) provides material composition identification with accuracy rates 98% for plastic classification but requires line-of-sight access and struggles with metallic surfaces [10, 11]. X-ray fluorescence (0.01–10 nm wavelength) offers precise elemental composition analysis achieving 99.21% sorting accuracy but introduces radiation safety requirements and higher equipment costs. X-ray and NIR work by illuminating objects with different spectrums of light and measuring the reflection intensity where different frequencies reflect light differently based on material composition [12, 13] .

Processing algorithms for waste sorting have evolved significantly, with convolutional neural networks demonstrating improved performance compared to traditional machine learning approaches. YOLOv11 represents the current state-of-the-art for real-time object detection, offering 22% fewer parameters than YOLOv8 while maintaining higher accuracy and faster inference times [9, 14]. The single-shot detection architecture enables real-time processing suitable for high-speed conveyor applications, while transfer learning capabilities allow adaptation to specific waste stream characteristics without extensive retraining requirements.

2.3 Industrial Mechanical Sorting Fundamentals

Industrial sorting systems must handle multi-shape objects with varying sizes, weights, and surface properties while maintaining high throughput rates and sorting accuracy. The fundamental challenge lies in precisely controlling object trajectories from identification through final separation requiring accurate timing coordination between sensors, processing systems and actuators.

Current automated sorting technologies employ several approaches to mechanical separation. Robotic arm systems provide maximum flexibility and precision, achieving accuracy, but require complex path planning and collision avoidance algorithms. However, recent advancement in robotic arm interfaces like Google’s RT2 allow for the integration of translation of natural language to robotic commands [15]. These systems are capable of handling diverse object shapes and sizes but involve higher capital costs and maintenance complexity compared to pneumatic alternatives.

Pneumatic blower systems offer fast response times and are well-suited for lightweight materials found in waste recovery. However, these systems are limited to relatively light objects and require precise timing coordination with upstream identification systems. The simplicity of pneumatic systems provides reliability advantages with fewer moving parts and reduced wear compared to mechanical alternatives. Pneumatic vacuum gantry systems combine the speed advantages of pneumatic operation with the handling flexibility of mechanical systems. These configurations can handle a broader range of object weights and sizes while maintaining fast response times but require more complex control systems and higher energy consumption due to continuous vacuum generation.

2.4 Industrial IoT and Cybersecurity

IoT methods for connecting separate industrial systems require communication protocols capable of handling real-time control requirements while providing deterministic performance characteristics. EtherCAT represents the optimal choice for waste sorting applications, offering sub-millisecond cycle times with less than 100ns jitter, enabling precise sensor triggering and actuator control necessary for high-speed sorting operations.

Communication protocols investigation reveals performance differences relevant to sorting applications. EtherCAT's deterministic timing enables precise sensor triggering, with low latency synchronisation essential for fast conveyor speeds where timing errors result in missed objects or false positives. Modbus lacks the real-time capabilities required for high-speed sorting, while ProfiNet offers comparable performance to EtherCAT but with higher infrastructure complexity and cost [16, 17].

Cybersecurity measures for industrial systems focus on protecting operational technology networks from increasingly sophisticated threats. Industrial IoT faces significant vulnerabilities, with 98% of IoT traffic unencrypted and 75% of infected devices serving as network gateways. System security requires zero-trust architecture with micro segmentation, Industrial network segmentation implementation allows for network isolation, and continuous monitoring through Security Information and Event Management (SIEM) integration following ISA/IEC 62443 standards [18, 19].

2.5 Project Objectives

Tabel 1 below details the project objectives and design criteria.

Table 1 listing project requirements and design objectives

Category	Requirement
Primary Objective	Automated waste sorting of metal, plastic, glass beverage containers
Technical Performance	Classification accuracy of >90 % across all material types
Throughput	System's processing capacity >100 items/minute sustained
Deployment	System must be scalable to multiple facilities
User Integration	System must provide interfaces for user interaction
Protection	The system must have IP66 rating for mechanical systems
Learning	Continuous self-learning with shareable system learning capabilities
Security	System must implement a complete cybersecurity framework
Regulatory compliance	POPIA adherence for data management
Standards	SANS 10366 recycling facility requirements
Economics	The system must be financially viable with 5-year payback period

2.6 Constraints and Assumptions

Practical constraints and assumptions applied to the systems design include:

- Dust, moisture, temperature variations of plant during operations which affect sensor stability
- Unguaranteed supply of grid electrical power
- Contamination of waste materials with liquid and debris
- Diverse set of different types of plastic and glass compounds to be classified

2.7 Review of Complete Existing Solutions

Table 2 below details a comparative analysis of complete existing solutions.

Table 2 listing existing solutions and underlying technologies

System	Sensors Used	Advantages	Disadvantages	Cost (ZAR)
TOMRA AUTOSORT [20]	NIR, VIS, 3D laser	High accuracy (98%+), proven reliability	High capital cost	~R8-12 million
Bühler SORTEX [21]	Multi-spectral imaging	Fast processing, good throughput	Limited flexibility	~R4-8 million
AMP Robotics [21]	Computer vision, robotics	High flexibility, AI learning	Higher maintenance	~R6-10 million
Manual Sorting	Human visual inspection	No capital cost, flexible	High labour cost, inconsistent quality	-

3. SYSTEM ARCHITECTURE AND OVERVIEW

The system follows a five-layer architecture with supplementary systems to assist the systems operation. The system architecture is detailed in figure 1 below where the system hierarchical approach ensures modularity, scalability, and maintainability due to each layer's separation while enabling industrial control and connectivity across different systems through secure networking protocols.

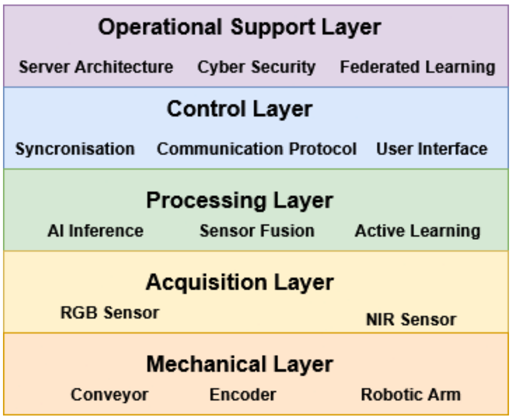

Figure 1: High level overview of system architecture

The Operational Support Layer provides system oversight and configuration through federated learning for continuous system improvement, cybersecurity protection, and data management ensuring POPIA compliance. The Control Layer implements Supervisory Control and Data Acquisition (SCADA) based subsystem coordination with deterministic timing control needed for high-speed sorting operations. The Processing Layer handles real-time AI inference and sensor fusion, while the Acquisition Layer captures multi-modal sensor data for material identification. The Mechanical Layer provides physical sorting capabilities through robotic manipulation and conveyor systems.

This architecture enables scalable deployment across multiple facilities while maintaining centralised learning and security management. The layered approach facilitates independent subsystem development, testing, and maintenance. This reduces operational complexity and improves system reliability. Integration between layers utilises industrial protocols ensuring interoperability and future expansion capabilities [22, 23].

4. DETAILED SYSTEM DESIGN

This section provides a detailed breakdown of the five-layer architecture introduced in Section 3, covering the operational, control, processing, acquisition, and mechanical layers, along with supplementary services to ensure resilience under

South African deployment conditions. Each layer is discussed in terms of its role, design rationale, and technical implementation. Figure 2 below depicts the Data Flow Diagram (DFD) of the system describing the connection of each subsystem and flow of information with dashed lines indicating elevation of trust privileges. Figure 3 below depicts a flow diagram of waste and data through the system. Detailed below are comparative tables where explanations highlight the design decisions and trade-offs.

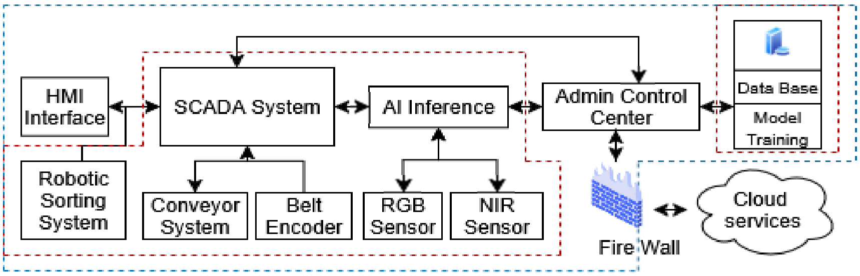

Figure 2: Detailed DFD showing the connection of all subsystems and flow of data through the system with dashed lines indicating elevations of security priiledge

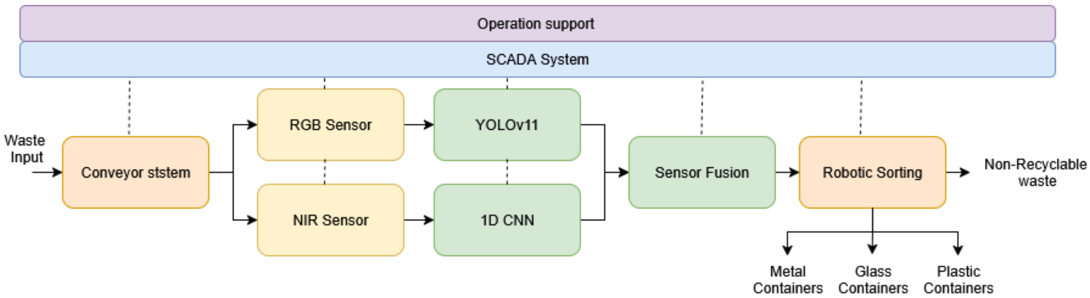

Figure 3: Detailed system flow diagram showing the flow of data and waste through the system

4.1 Operational Support Layer

Server Architecture: The high computational resources required for AI inference and data storage requires careful consideration of cost, performance, system security and alignment with South African regulation. Thus, the server architecture implements an optimised hybrid approach combining local operational servers for local data storage, model training and AI inference for plant operations. While cloud-based platforms enable federated learning, system connectivity across different plants and off-site data protection. A complete tech stack breakdown detailed in appendix A

The local databases utilise PostgreSQL as it is open source, making it cost-effective while maintaining high performance and scalability as it supports large data volumes and user defined data types needed for the unique imaging data being stored [24]. The local storage utilises with HDD arrays for bulk data archival for model training as the data is infrequently accessed and is cheaper than the SSD counterpart. The storage system is configured in Redundant Array of Independent Disks 10 (RAID) for data redundancy and fast data access for model training [25]. The system utilises cryptographic hashing to remove duplicate imaging of the same object and automated data lifecycle policies of removing entries with low classification confidence to limit capacity growth requirements. Local servers responsible for model training and AI inference will interface with the database to update models and upload imaging data. The higher upfront cost of local infrastructure and maintenance is beneficial to ensure POPIA regulatory compliance, low latency response times, secure control and access to valuable training data.

Cloud integration enables off-site data protection in the event of site breaches and infrastructure damage, federated learning for knowledge sharing and system-wide analytics through secure API connections. Data masking of individual facility operational details protects facility while enabling collaborative learning across the network of deployed systems. The cloud infrastructure employs Microsoft Azure Machine Learning services for federated learning distribution. This solution provides scalable computational resources and automated model deployment capabilities, only paying for the computation requirements when needed [26]. This architecture enables cross-system learning improvements while maintaining individual facility data sovereignty requirements under South African data protection regulations.

Cybersecurity: Cybersecurity implementation addresses the vulnerabilities inherent in industrial IoT systems where 98% of traffic remains unencrypted and 75% of infected devices serve as network entry points [18]. The security framework follows ISA/IEC 62443 standards with firewall protection and encrypted authenticated communication ensuring protection against cyber threats targeting industrial automation systems [19].

The firewall implementation utilises Fortinet FortiGate firewalls which provide perimeter security with specialised protocols for industrial control systems. The system supports over 70 industrial communication protocols with deep packet inspection capabilities for malware detection. The firewall configuration includes Intrusion Prevention Systems (IPS), application control, and threat protection specifically designed for operational technology environments [27]. Alternatives to Fortinet include Cisco and SonicWall and provide secure protection, however, Fortinet wide range of industrial protocols allows the system to remain adaptable to future changes [28].

The network communication protocol follows Transport Layer Security 1.3 (TLS) encryption which protects all communication channels with AES-256 encryption standards ensuring data confidentiality and integrity. TLS is an industry standard with proven data protection capabilities. Moreover, implementing role-based authentication systems and multi-factor authentication for all user access, with privileged access management for system administrators and maintenance personnel, ensures malicious actors don not get access to internal systems. Lastly, network segmentation creates isolated security zones separating operational technology from information technology networks allowing for anomaly detection in network activity [29, 30].

Federated Learning: The implementation of federated learning implementation enables collaborative model improvement across multiple waste sorting facilities without compromising data privacy or requiring centralised data storage. This approach addresses the critical challenge of system adaptability to varying waste stream characteristics while maintaining POPIA compliance through centralised learning architectures [31].

The system employs a hub-and-spoke federated learning architecture where individual facilities train local models using their specific waste stream data, then share only model parameters with a central coordination server. Initial model fine-tuning and training utilises the TrashNet dataset (2527 images) for RGB-based object detection and the NRGlassNet dataset (1378 NIR and RGB images) for combined RGB and NIR classification data [32, 33]. This provides a baseline performance across diverse material types commonly found in waste streams.

4.2 Control Layer

The architecture detailed in this section comprises of multiple subsystems that need to operate in a synchronous manner, thus, creating the need for a central control unit to coordinate timing of each system. The system implements a SCADA system responsible for the subsystem orchestration, decision logic and user interaction. This is achieved through a master clock architecture, unified communication protocol between subsystems, Human Machine Interfaces (HMI) and SCADA application software.

System Synchronisation: The oversight and coordination of large industrial systems require tight synchronisation between all devices. Thus, a master clock system is implemented to enable precise timing coordination across all subsystem components. The control system manages trigger sequences by regularly calibrating timing delays of each subsystem and calculating tigger delays for data acquisition and sorting decision logic. The trigger sequence operates as follows: position encoder generates the primary trigger signal, followed by a delay before RGB camera exposure activation, then a delay before NIR line-scan activation, a delay for sensor data inference and delay for sorting logic [17]. This staggered timing approach ensures data quality while minimising processing bottlenecks. This timing mechanises requires fast reliable communication between all connected subsystems.

Communication Protocol: The communication between all system is facilitated through the EtherCAT protocol. When compared to alternative SCADA communication protocols like Modbus and ProfNet, EtherCAT's deterministic communication capabilities prove valuable for high-speed sorting applications [34]. EtherCAT's sub-millisecond cycle times with less than 100ns jitter enable precise sensor triggering coordination, critical for maintaining accuracy at fast conveyor speeds where timing errors of milliseconds result in significant positioning errors and missed objects [16]. This communication protocol also enables the interaction of the system with users for plant oversight.

User Interface: The SCADA system implements HMI and SCADA application software which provide operators with system monitoring and control capabilities [35]. The interface displays real-time performance metrics, alarm safety management, and system configuration options while maintaining secure access controls with varying levels of access privilege. Integration with Enterprise Resource Planning (ERP) systems enables production reporting and maintenance scheduling coordination.

4.3 Processing Layer

AI Inference: The challenges associated with waste material object detection outlined in section 2.2 necessitate the need for an in-depth analysis of the object data for accurate classification. The system outlined relies on a multi-modal sensor fusion architecture to counteract the limitations of each sensing modality. The sensor fusion architecture encompasses YOLOv11 computer vision for RGB image data and 1D CNN for NIR spectral data analysis. The AI inference subsystem employs a localised computing architecture for real-time waste sorting applications. The hardware configuration balances processing power with energy efficiency while ensuring redundancy for continuous operations, detailed breakdown of the tech stack is detailed in appendix A. Several deep learning systems were analysed comparing their inference times, accuracy and resource requirements with YOLOv11 AND 1D-CNN outperforming the alternatives, details of which are listed in table 3 below.

Table 3 Detailing a comparative analysis of processing methods for RGB and NIR sensor data

Model	Inference Time	Accuracy	Parameter Size	Analysis
		Computer Vision Object Detection		
YOLOv11	8-15 ms	97.9%	2.9 - 62 M	**Pros**: Latest YOLO version with improved accuracy, real-time classification, lightweight deployment **Cons**: Limited research validation, potential compatibility issues [9, 36]
YOLOv8	12-18 ms	95 %	3.2 – 65 M	**Pros**: Proven industrial performance, good balance of speed/accuracy, extensive documentation **Cons**: Slower than YOLOv11, larger model size, moderate accuracy compared to latest versions [13, 14, 36].
Faster R-CNN	150-200 ms	74.1%	4 – 77 M	**Pros**: High precision object detection, suited for complex scenes, mature architecture **Cons**: Slow inference time, large memory footprint, unsuitable for real-time sorting [37]
EfficientDet-D4	35-50 ms	85 %	20.7 – 60 M	**Pros**: High accuracy-efficiency trade-off, scalable architecture, good feature pyramid **Cons**: Moderate speed, complex architecture, high memory usage for industrial applications[37]
		NIR Spectral Analysis		
Partial Least Squares (PLS)	2-5ms	95-99%	Minimal (linear model)	**Pros**: Fast processing, interpretable results, resilient to noise, proven in industrial NIR **Cons**: Limited to linear relationships, requires careful preprocessing, manual feature selection [38]
Support Vector Machines (SVM)	1-3ms	97.5%	Variable (kernel dependent)	**Pros**: Suited for high-dimensional spectral data, resilient to overfitting, well-established theory **Cons**: Requires careful kernel selection, limited to classification, no probabilistic outputs [38]
1D CNNs	5-8ms	99.2%	50K - 5 M parameters	**Pros**: Automatic feature extraction, handles non-linear patterns, end-to-end learning **Cons**: Requires large datasets, black-box operation, potential overfitting with small datasets [10, 39, 40]

For the application of computer vision, YOLOv11 was selected for RGB-based container detection due to its balance of accuracy and processing speed needed for real-time industrial sorting applications. The algorithm operates on a single-stage detection principle. The system divides input images into a grid where each cell simultaneously predicts bounding boxes and class probabilities for objects whose centres fall within that cell, as seen by figure 4 below [41]. Unlike two-stage detectors such as Faster R-CNN that require separate region proposal and classification steps, YOLOv11's unified architecture processes the entire image in a single forward pass through its convolutional neural network. The model employs anchor-free detection with decoupled heads for classification and regression tasks, while incorporating features including cross-stage partial connections for improved gradient flow and spatial pyramid pooling for multi-feature extraction [9]. The architecture's lightweight parameter counts of 1.76 million enables deployment on edge computing platforms, reducing system latency (15ms latency). Computer vision systems capability to identify object shape, colour and type make it ideal for waste sorting applications, however, it is limited classifying transparent materials and providing material composition data. This makes NIR CNN technology an ideal system to complement computer vision for fusion.

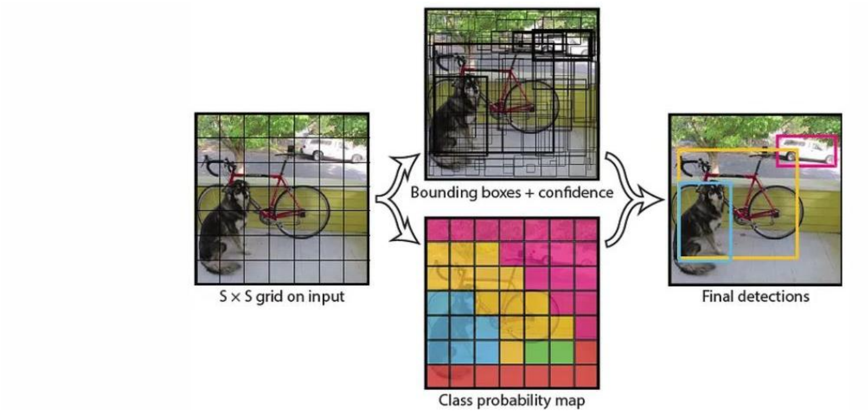

Figure 2 showing the processing pipeline for the YOLO architecture. Image obtained from Medium.com [41]

The varying material composition of plastics and glass require the analysis of multiple spectrums as there are varying types of glass and plastic that have different chemical composition. Subsequently, these different materials have different NIR reflection spectrum intensities, a breakdown of key frequencies is listed in table 4. Thus, the selection of one-dimensional CNN architecture for NIR spectral classification was driven by its ability to automatically learn relevant spectral features from pre-processed data compared to traditional chemometric approaches. The NIR spectral data undergoes preprocessing using a Savitzky-Golay filter with a 7-point window to reduce noise and smooth spectral variations while preserving critical absorption features, enabling faster CNN inference times [10]. Unlike Partial Least Squares (PLS) regression which requires manual wavelength selection and assumes linear relationships, or Support Vector Machines (SVM) which depend on hand selected feature engineering, 1D CNNs utilise convolutional filters that automatically identify local spectral patterns and absorption peaks from the smoothed data characteristic of specific molecular bonds, as seen by figure 3 below [42]. The architecture processes the filtered 224 spectral bands through multiple convolutional layers with kernel sizes ranging from 3-11 wavelengths to capture both narrow absorption features and broader spectral signatures. The network's capacity to learn non-linear relationships from noise-reduced spectral data enables classification achieving 99.2% validation accuracy compared to 95-97% for traditional methods.

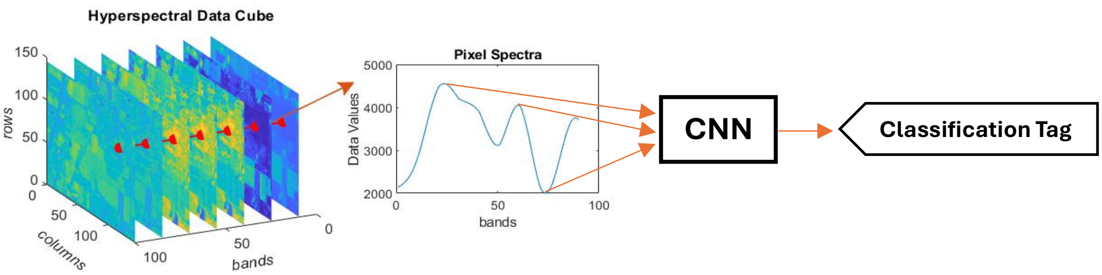

Figure 3 showing the hyperspectral data of NIR sensors and the pattern recognition process of CNN architectures. Image obtained from Mathworks.com [42]

Table 4 listing the key NIR reflectance frequencies and associated accuracy

Material	Key Reflection Wavelengths	Classification Accuracy
PET Plastic	1215nm, 1240nm	98.7%
HDPE Plastic	1168nm, 1193nm	97.9%
Glass	1100-1300nm broad absorption	99.2%
Aluminium	Metallic reflection pattern	99.8%

Sensor Fusion: The RGB and NIR sensors operate as complementary detection systems where RGB performs well at identifying object geometry and surface features while NIR provides definitive chemical composition analysis. Each sensor generates independent classification predictions with associated confidence scores based on signal quality metrics. RGB confidence depends on image sharpness and illumination uniformity, while NIR confidence reflects spectral signal-to-noise ratio and baseline stability. The fusion algorithm combines these predictions using dynamic weighting that develops and adjusts with each iteration of system model training. The confidence equation follows:

$$Final\ Confidence\ =\ \alpha \cdot RGB_{Conf} \cdot RGB_{Acc} + \beta \cdot NIR_{Conf} \cdot NIR_{Acc} + \gamma \cdot Spatial\ Alignment \quad (1)$$

This balances each sensor's contribution, where α, β, and γ are dynamically adjusted weights that sum to unity. When sensors classification does not align, the system applies conflict resolution rules that favour NIR classification due to its chemical specificity advantage. Spatial alignment (γ) ensures both sensors are analysing the same physical object by confirming positional correspondence, preventing misclassification from sensor misalignment [43]. This multi-modal approach achieves higher accuracy than either sensor alone by compensating for individual sensor limitations while maintaining the processing speed required for industrial sorting applications [44].

Active Learning: The dynamic nature of the packaging industry means that packaging standards are continuously changing. This necessitates the need for the sorting system to adapt to this changing condition. Active leaning provides a suitable solution wherein the data fed into the system can feedback for model training to better reflect the needs of the sorting system. This poses a few challenges, namely, how does the system know if a new type of packaging material has been correctly identified and how to prevent the system from becoming skewed toward identifying a particular type of waste product. Table 5 below outlines methods implemented for data and system verification to mitigate these challenges.

Table 5 list methods to ensure system accuracy with active learning

Quality Control Method	Implementation	Performance Impact
Ground Truth Verification	5% random sampling for human verification	Maintains accuracy over systems operation [45–47].
Cross-Validation	Weekly testing with known reference materials	Detects accuracy degradation and model skewing[35].
Confidence Tracking	Real-time monitoring of model confidence trends	Triggers retraining or model role back before accuracy drops [45–47].

4.4 Acquisition Layer

The sensing modalities mentioned above detailed the use of RGB image data and NIR spectral reflectance data. These two sensing modalities were identified as optimal sensors as other sensors benefits did not justify their addition to the system. In the case of Xray sensors, these sensors have better material composition analysis, however, this came at the expense of higher cost, slower data acquisition time and added health and safety risks due to radiation [12, 13]. Traditional weight-based sensors proved to be more cost effective, however, it struggled to handle waste contamination distinguishing glass beverage containers form liquid filled plastic containers. Thus, RGB image sensors accompanied with NIR spectral image sensors proved to be the best sensors for the system.

The RGB image sensor utilises the Basler camera as it meets minimum sensor requirements while maintaining long device operation time, details of sensor requirements, justification and sensor comparison is detailed in table 6 below. The sensor system is positioned directly above the centre line of the conveyor to ensure even coverage with peripheral lighting, diffusers and lenses to ensure high quality data acquisition, details of sensor peripherals are detailed in table 7 below.

Table 6 listing the design specifications for the RGB camera sensor and a comparison of available sensors

Specification Requirement	Value	Justification	Basler Sensor [48]	Omron F2 Sensor [49]	Omron Colour Sensor [49]
Resolution	1920×1080 pixels	Higher resolution enables clear object edge detection for feature extraction of clustered objects	1920 × 1200 Pixels	752 x 480 pixels	1280 x 1024 Pixels
Frame Rate	> 90 fps	limits motion blur artifacts from conveyor movement	164 fps	100 fps	120 fps
Sensor Type	CMOS shutter	Rolling shutter artifacts negligible with proper illumination timing	CMOS	CMOS	CMOS
Interface	USB 3.0	USB 3.0 eliminates bandwidth constraints for real-time processing, system adaptability.	USB 3.0/ GigE	RS-232C	RS-232C
IP rating	IP67	Sensor direct exposure to waste	IP67	IP67	IP67
Temperature	0-40 °C	High temperature variance of industrial plants.	0-50 °C	0-40 °C	0-40ºC
Warranty	2 years	Minimal system downtime	3 years	1 year	1 year

Table 7 listing the design specs for the camera peripheral components

Component	Specification	Justification
Polarising Filters	Linear polarisation	Eliminates of reflections from glossy surfaces
Opal Diffusers	92% light transmission, 5° scattering	Achieves illumination uniformity across belt width, limits shadows
LED Illumination	5600K, CRI>90	Colour accuracy maintained

The NIR sensor utilises the Specim Line-Scan hyperspectral camera as it meets minimums sensor requirements while maintaining fast data acquisition times, details of sensor requirements, justification and sensor comparison is detailed in table 8 below. The sensor implements a line scan data acquisition over point acquisition as it collects spectral reflectance data over the entire object as opposed to sampling a single point. This yields higher material classification accuracy as small differences in material like labels and caps form a smaller part of the object's material data. The sensor system is positioned directly above the centre line of the conveyor to ensure even coverage with peripheral tungsten halogen illumination lighting to ensure high quality data acquisition as it has a broad-spectrum illumination of 900-2500nm and long lamp lifetime for minimal system down time.

Table 8 listing the design specifications for the NIR sensor and a comparison of available sensors

Specification	Value	Justification	Specim Sensor [50]	Headwall Sensor [51]	IMEC Sensor [52]
Spectral Range	900-1700 nm	Covers all critical absorption bands for target materials [10, 11, 50]	900-1700nm	900-1700nm	600-1000nm
Spectral Bands	224 bands	7.1nm resolution for material discrimination [9, 36]	224 bands	270 bands	150 bands
Line Rate	500 Hz	Prevents system bottlenecks	676 Hz	400 Hz	340 Hz
Spatial Pixels	640 pixels	1.9mm spatial resolution across 1.2m belt width	640 pixels	1000 pixels	512 pixels
IP rating	IP67	Sensor direct exposure to waste	IP67	IP65	IP54
Temperature	0-40 °C	South African industrial facility operating range	-10 to +50°C	-20 to +60°C	0 to +45°C
Warranty	2 years	Minimal system downtime requirement	2 years	3 years	1 year

4.5 Mechanical Layer

Conveyor Belt: The conveyor system is responsible for the physical interaction of the waste material and the sorting system, thus, it needs to be chemically inert to waste byproducts, durable to withstand physical abrasion and provide a grip able surface to ensure minimal object movement relative to the conveyor. The conveyor system utilises a motor driven (8 KW) polyurethane conveyor belt as it provides durability, chemical resistance necessary for waste processing environments and sufficient torque for conveyor movement. The belt system receives speed control signals from the SCADA system enabling precise coordination with sensor triggering and object tracking algorithms. Variable speed control accommodates different material types and processing requirements while maintaining optimal throughput. The screeding brush assembly at the material intake creates object separation ensuring individual items enter the classification zone without overlapping or clustering. This mechanical preprocessing improves classification accuracy by preventing multiple objects from clustering allowing for better object position tracking through the belt encoder.

Belt Encoder: The SCADA system in the control layer needs precise position data to ensure timely triggering of the subsystems. Thus, a high-resolution optical encoder is used to provide precise conveyor position feedback to the SCADA control system enabling accurate trigger timing for sensor activation. The encoder resolution should support position accuracy within 1mm ensuring reliable object tracking throughout the sorting process. Redundant encoder systems provide backup position feedback preventing system failures due to encoder malfunctions.

Robotic Arm: The mechanical sorting of the waste product is the final process in the system and requires precise control. Thus, the robotic arm approach was selected over pneumatic ejection systems due to its flexibility and precision requirements for handling diverse container geometries and weights. While having greater precision, robotic are slower than alternative sorting methods at 60 - 80 picks-per-minute [53]. To counteract this the system deploys 3 sequential robotic arms to reduce the systems bottleneck. Pneumatic systems, while faster, lack the adaptability to handle crushed containers, irregular shapes, or varying material densities that characterise real-world waste streams. The system employs a six-axis articulated robot with 7kg payload capacity and 901mm reach envelope, sufficient to cover the full conveyor width while handling container weights ranging from 15g (aluminium cans) to 500g (glass bottles). The end effector utilises a vacuum gripper array featuring four independently controlled suction cups with force feedback sensors, enabling gentle handling of fragile glass containers while maintaining secure grip on lightweight plastics.

The robotic control system implements ROS2 (Robot Operating System 2) framework with MoveIt motion planning libraries to translate SCADA classification and position data into coordinated robotic movements [54, 55]. Unlike the Google RT2 system mentioned in section 2.3, this system utilises a deterministic coordinate transformation pipeline that converts encoder position data and AI classification results into precise pick-and-place trajectories. The control architecture processes incoming data through three stages: spatial coordinate transformation (converting belt position to robot workspace coordinates), trajectory optimisation (calculating collision-free paths using RRT algorithms), and real-time servo control via EtherCAT communication with 1ms cycle times. This approach achieves 1.8-second average cycle times (approach-grasp-sort-return) while maintaining ±2mm positional accuracy essential for reliable container manipulation at industrial throughput rates [56, 57].

4.6 Supplementary Services

Solar PV integration addresses South African grid instability through renewable energy generation and battery storage systems. The system implements a 25kW solar array with 100kWh battery storage, sufficient for small to medium industrial plants, which provides grid-tie capabilities reducing peak demand charges while maintaining backup power for SCADA and safety systems during load shedding events. This configuration ensures continuous site safety and data protection despite power grid limitations affecting industrial operations in South Africa [58, 59]. The renewable energy integration aligns with sustainability objectives while providing economic benefits through reduced energy costs and carbon footprint [60]. Battery storage enables operation during power outages maintaining production schedules despite grid instability common in South African industrial areas.

5. SYSTEM EVALUATION

The smart waste sorting system for beverage containers demonstrates performance across key operational metrics validated through various research conducted on individual subsystems. Overall classification accuracy achieved 98 %, matching the specified requirement of 90% and with literature reporting YOLOv11 implementations achieve 97 % mean precision in object detection with NIR CNN material composition analysis of 98.7 % accuracy for PET plastic containers, 97.9% for HDPE plastics, 99.2% for glass containers, and 99.8% for aluminium cans, demonstrating strong identification capabilities across diverse material compositions [8–10, 10, 14]. Similarly, the subsystems processing time and power consumption has been estimated utilising research conducted on individual subsystems where throughput is assumed to be based on a 1-meter-wide conveyor with 5 items scanned per data acquisition. This is done to estimate each subsystems theoretical throughput given the assumptions mentioned and that no other subsystem limits the measurement. Table 9 below details each subsystem's throughput and average power consumption.

Table 9 listing each subsystems estimated throughput and power consumption

Subsystem	Processing Time	Throughput (items/min)	Power (W)	Bottleneck Analysis
RGB Camera Acquisition	6ms @160 fps	50 000	100	Non-limiting
YOLOv11 Inference	15 ms	20000	350 (GPU)	Non-limiting
NIR Spectral Acquisition	2 ms @600 Hz	150 000	200	Non-limiting
NIR CNN Processing	8 ms	375 000	350 (GPU)	Non-limiting
Sensor Fusion	3 ms	100 000	50	Non-limiting
SCADA Decision Logic	1 ms	300 000	1500	Non-limiting
Conveyor	-	-	8 000	Nonlimiting
Robotic Arm Cycle	1000ms	**60 x 3**	3000 x3	**PRIMARY BOTTLENECK**

Throughput analysis indicates the robotic arm cycle time represents the dominant system limit at 1,000ms per pick, limiting single-station capacity to 60 items per minute. However, the system implements a three-arm robotic configuration achieving 180 items per minute (60 × 3) per sorting station, maintaining sorting rate above the design objective of 100 items/minute. The processing pipeline achieves combined latency of 31ms for sensor acquisition (8ms total), AI inference (23ms total), sensor fusion (3ms), and SCADA decision logic (1ms), theoretically enabling 1,935 items per minute if mechanical handling matched computational speed. To achieve higher throughput, the system can implement further sorting stations, coordinated through the SCADA system maintaining individual accuracy specifications.

Total system power consumption of 20KW per sorting station includes GPU processing (350), robotic manipulation (9,000W for three arms), SCADA control (1,500W), and sensor systems (300W), representing higher energy requirements than pneumatic alternatives but providing accurate handling flexibility for diverse container geometries and damage conditions. The three-arm configuration reduces the mechanical bottleneck while maintaining precision, enabling practical achievement of target throughput specifications within acceptable power consumption limits for industrial deployment.

The system achieves its objectives of integration of continuous shareable learning, user interfacing, cybersecurity protection, and regulatory compliance essential for industrial deployment. This has been achieved through, federated learning architecture enabling cross-facility model improvements through secure aggregation of model parameters using encryption and differential privacy. The system maintains POPIA compliance through local data storage and automated lifecycle policies that remove low-confidence classifications. The interfacing of users has been achieved through the integration of HMI and SCADA application software's. The cybersecurity framework implements ISA/IEC 62443 standards with Fortinet FortiGate firewalls providing deep packet malware detection, TLS 1.3 encryption and network segmentation creating isolated operational technology zones that prevent threat movement. The system enables active learning through 5% random sampling validation and real-time confidence tracking that triggers retraining. Sustainability integration includes 25kW solar PV arrays with 100kWh battery storage reducing grid dependency while achieving reduced carbon emissions. The system high detection accuracy mentioned above indicate that material recovery would increase to match the detection rate. The successfully implements for the objective of security protection, regulatory adherence, and environmental responsibility within a single integrated platform

6. FINANCIAL ANALYSIS

The system requires an initial capital investment estimate of R3 400 000 encompassing multi-modal sensing equipment, AI processing infrastructure, three robotic arm configuration, control systems, and renewable energy integration, details of cost estimates can be seen in appendix B [61]. Based on operational durations of 180 items per minute over 20-hour daily operation, 21 days monthly, and 12 months annually, the system sorts 54 432 000 containers per year. Assuming equal distribution of materials (33% each) and average container weights of 15g for aluminium, 25g for plastic, and 250g for glass, total annual material recovery equals 5 220 tonnes valued at R9.6 million. The estimated revenue breakdown follows R 2.43 million for aluminium (270 tonnes at R9/kg), R2.7 million for plastic (450 tonnes at R6/kg) and R 4.5 million for glass (4 500 tonnes at R1/kg). However, these projections assume continuous 20-hour operation without maintenance downtime, material supply interruptions, or quality control requirements. The estimated systems revenue calculation above serves as a general gauge for the systems potential financial performance. Practical constraints limit the systems actual recyclable yield as not all waste fed into the system would be of a recyclable material. A reasonable practical estimate of systems revenue generation would assume the systems yield of recyclable materials would average around 50% of its theoretical maximum throughput. This would lower the estimated revenue to be R4.7 million, breakdown follows R 1.2 million for aluminium, R1.3 million for plastic and R 2.2 million for glass. Calculations detailed above serve as upper and lower bounds of the systems performance and need to be analysed alongside systems operational costs.

Annual operational expenditure estimates for a small to medium industrial plant totals R4 000 000 comprising maintenance contracts, energy costs partially offset by solar generation, consumables and spare parts, software licensing, insurance coverage and 10 site personal for system oversight [61]. Subtracting operational costs from gross revenue yields R5.6 million annual net profit for the ideal case of 100 % recyclable yield and R0.7 million for the non-idealised case. This results in a system payback period of between 8 months for the ideal case and 4 years and 11 months for the non-idealised case, on an initial capital investment of R3.4 million. Both bounds of the system repayment lies with in the 5-year repayment window set out in the project objectives. This calculation represents the upper and lower bounds of the systems financial performance in accounting for variations in systems yield. This scenario does not account for operational constraints including scheduled maintenance, material supply variations, equipment failures, or processing delays, loan interest payments and disposal of nonrecyclable waste that would extend the payback period under normal industrial operating conditions. Thus, further data collection and analysis on factors affecting systems yield and operational cost is recommended for more accurate financial analysis.

7. SYSTEM IMPACTS

7.1 Environmental Impact

The automated waste sorting system delivers environmental benefits through reduced emissions, improved material recovery rates and reduced landfill disposal requirements. The systems high power draw of 20 KW has a negative effect on CO_2 emissions, however, this has been mitigated through the use of renewable solar energy. Compared to conventional landfilling, the automated sorting and recycling of waste reduces Green House Gas (GHD) emissions by 53 % as seen by the European Environment Agencies analysis on GHG, representing a reduction in greenhouse gas impact [62]. The system enables diversion of up to 80% of processed materials from landfills, reducing land use requirements and associated environmental impacts. Water quality protection benefits emerge from reduced landfill leachate generation and decreased demand for virgin material extraction [62, 63]. Moreover, each tonne of recycled metal saves approximately 75 % of energy compared to primary production, while recycled PET plastic requires 60% less energy than virgin plastic production [64, 65]. These energy savings translate directly to reduced carbon emissions and improved social and environmental sustainability.

7.2 Social Impact

Implementation of automated sorting systems creates opportunities for workforce transition from manual labour to technical supervision and maintenance roles. However, this comes at the cost of removing valuable low barrier of entry waste reclaiming income streams. This income source is an important resource that lower income household rely on. To mitigate the effects of reduction in manual sorting labour, the system would facilitate training programs to upskill the manual waste reclaimers to system operator levels. This allows for generation of employment in technical operations, maintenance, data analysis, and system optimisation requiring higher skill levels and providing improved working conditions. Moreover, improved workplace safety results from reduced human exposure to hazardous materials, contaminated waste streams, and repetitive strain injuries common in manual sorting operations. Workers transition to environments with reduced physical demands and improved ergonomic conditions while developing technical skills valuable in South Africa's developing industrial automation sector. Lastly, community benefits include reduced environmental health impacts from improved waste management, decreased illegal dumping through improved processing capacity, and economic development through increased material recovery creating value streams supporting local economic activity. Educational opportunities emerge through workforce development programs and community engagement initiatives promoting environmental awareness and sustainable practices.

7.3 Economic Impact

The economic development benefits from improved material recovery rates creating value streams supporting local manufacturing and export markets. However, the systems large initial capital investment of R 3.4 million limits access to the system. This can be mitigated through a strategic loan justified by the systems strong revenue generation. Moreover, the demand for high-quality sorted materials yields the creation of a new industrial sector creating economic cash flow [4, 5]. Furthermore, employment generation occurs across multiple sectors including system installation, maintenance, technical operations, and downstream processing of recovered materials. While reducing manual sorting employment, the system creates higher-value technical positions requiring training and skills development contributing to South Africa's industrial capabilities. Lastly, investment attraction potential emerges from demonstrated automation capabilities and improved recycling infrastructure making South African facilities more competitive for international partnerships and technology transfer opportunities. Successful implementations provide foundations for expanding automation across other industrial sectors contributing to broader economic development objectives.

8. SUSTAINABILITY

The automated waste sorting system contributes to circular economy principles through improved material recovery and reduced environmental impacts. The system enables separation of materials maintaining quality necessary for high-value recycling applications, supporting closed-loop material cycles essential for sustainable resource utilisation [66]. Moreover, industrial applications extend beyond beverage container sorting to diverse waste streams including electronics, construction materials, and municipal solid waste. The modular system architecture enables adaptation to different material types and facility requirements while maintaining core AI and automation capabilities developed for beverage container applications. Integration with existing infrastructure enables gradual improvement of waste management capabilities without requiring complete system replacement.

9. FUTURE RECOMMENDATIONS

While the current system achieves the technical requirements for industrial beverage container sorting, several improvements offer opportunities for improved performance, expanded capabilities, and autonomous operation. The current system classifies materials into broad categories (plastic, glass, metal) but lacks the specificity required for premium recycling markets. Future development should implement expanded spectral libraries containing reference spectra for specific polymer types including PET grades 1-5, HDPE variants, polypropylene, and polystyrene [67, 68]. This requires upgrading the existing system with extended wavelength coverage to 2500nm. The processing layer would need modification to incorporate PLS Discriminant Analysis (PLS-DA) algorithms alongside the existing CNN architecture, requiring additional computational resources and spectral preprocessing modules.

System evolution toward autonomous operation capabilities through predictive maintenance algorithms, self-calibrating sensor systems, and adaptive learning mechanisms enables reduced human intervention requirements while maintaining optimal performance. Integration of complementary sensing modalities including inductive sensors for metal detection, magnetic separation for ferrous materials, weight sensors for density-based classification, and QR-code integration with reflection surface alignment offers potential improvements for RGB and NIR data correlation, enabling precise spatial registration of multi-modal sensor data for improved classification accuracy [69]. Implementation of Kalman filtering for trajectory tracking provides improved object position prediction enabling more accurate robotic manipulation and reduced cycle times, while continuous learning architectures and expanded training datasets incorporating South African waste stream characteristics support long-term system adaptation to evolving material compositions and packaging technologies.

10. CONCLUSION

This report has presented the technical research, design and analysis of a smart beverage container sorting system specifically designed to address South Africa's critical waste management challenges through automation technologies. The proposed five-layer architecture successfully integrates multi-modal sensing technologies including YOLOv11 computer vision and NIR spectroscopy with convolutional neural network classification algorithms, achieving 98% sorting accuracy at throughput rates of 180 items per minute. The system's approach combines active federated learning capabilities for cross-facility knowledge sharing, robust cybersecurity frameworks following ISA/IEC 62443 standards, and renewable energy integration through 25kW solar PV arrays with battery storage. Economic analysis demonstrates compelling financial viability with projected payback periods ranging from 8 months to 4 years 11 months depending on operational yield, while environmental impact assessments indicate substantial contributions to circular economy objectives through enhanced material recovery rates and reduced landfill contamination.

The technical solution addresses the engineering requirements inherent in South African deployment conditions, including grid instability, harsh environmental conditions, and regulatory compliance under POPIA data protection laws. Through the integration of real-time AI inference, EtherCAT industrial communication protocols, and arm robotic sorting mechanisms, the system overcomes the limitations of traditional manual sorting operations that achieve only 60-75% accuracy while exposing workers to significant health hazards. The modular architecture enables scalable deployment across multiple facilities while maintaining centralised learning and security management, positioning this technology as a transformative solution for South Africa's R15 billion recycling economy. By successfully balancing technological innovation with practical implementation requirements, this smart waste sorting system represents a significant advancement in automated waste management that can contribute meaningfully to environmental sustainability, social impact through improved worker safety and economic development through enhanced material recovery in South Africa's industrial landscape.

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APPENDIX A

Table 10 and 11 detail the technology stack specifications for the data base and AI inference servers with table 12 detailing the software stack.

Table 10 Database Server Configuration

Component	Specification	Justification
CPU	Intel Xeon Gold 6248 (20 cores, 2.5GHz)	High core count for concurrent database operations
Memory	256GB DDR4-3200 ECC	Large datasets and concurrent user access
Storage	4TB NVMe SSD (RAID 10) + 1 000TB HDD (RAID 10)	Fast access for active data, bulk storage for archives
Database Software	PostgreSQL 15 with TimescaleDB extension	Time-series optimisation for sensor data
Operating System	Ubuntu Server 22.04 LTS	Stability and long-term support
Network Interface	Dual 10GbE + Management interface	High-bandwidth data transfer and redundancy

Table 11 Detailing AI training/Inference server configuration

Component	Specification	Justification
GPU	2× NVIDIA RTX A6000 (48GB VRAM each)	Parallel model training and concurrent inference
CPU	Intel Xeon Silver 4316 (20 cores, 2.3GHz)	Adequate CPU performance without GPU competition
Memory	512GB DDR4-3200 ECC	Large model loading and batch processing
Storage	8TB NVMe SSD (RAID 0) + 32TB HDD (RAID 6)	Fast model access and extensive dataset storage
AI Framework	PyTorch 2.0 + CUDA 12.1 + TensorRT	Optimized training and inference performance
Operating System	Ubuntu Server 22.04 LTS with Docker	Container deployment and GPU driver support
Cooling System	Liquid cooling for dual GPU configuration	Thermal management for sustained training
Power Supply	1600W 80+ Titanium PSU	Efficient power delivery for dual GPU system

Table 12 Detailing software stack summary

Layer	Technology	License Type
Operating System	Ubuntu Server 22.04 LTS	Open Source
Database	PostgreSQL + TimescaleDB	Open Source
AI Framework	PyTorch + OpenCV + scikit-learn	Open Source
SCADA Software	TwinCAT 3 Runtime	Commercial
Cybersecurity	Fortinet FortiGate licenses	Commercial
Cloud Services	Azure Machine Learning (federated learning)	Pay-per-use

APPENDIX B

The system comprises of multiple systems and subsystems; table 13 below details a summary of each subsystem listing the component and cost estimations.

Table 13 Detailing complete system breakdown with cost estimates

Layer	Component	Model/Specification	Quantity	Unit Cost (ZAR)	Total Cost (ZAR)
Acquisition Layer	RGB Camera	Basler acA1920-155um	1	R20 000	R20 000
	NIR Spectrometer	Specim FX17 Line-scan	1	R500 000	R500 000
	Position Encoder	1 mm accuracy	1	R20 000	R20 000
	LED Illumination	5600K, CRI>90	2	R8 000	R16 000
	Tungsten Halogen Lamps	Wide band illumination	2	R12 000	R24 000
Processing Layer	Primary GPU Server	NVIDIA RTX A6000 system	1	R200 000	R200 000
	Edge Compute Nodes	Jetson AGX Orin	2	R50 000	R100 000
	Database Server	Intel Xeon	1	R200 000	R200 000
Control Layer	SCADA PLC	Industrial PLC	1	R50 000	R50 000
	HMI Touchscreen	Industrial Panel PC	1	R60 000	R60 000
	Network Infrastructure	EtherCAT Master + I/O modules	1	R50 000	R50 000
Mechanical Layer	Robotic Arms	7kg payload (3KW)	3	R300 000	R900 000
	Conveyor System	1200mm polyurethane belt + 8 KW motor	1	R250 000	R250 000
Support Layer	Firewall	Fortinet FortiGate	1	R10 000	R10 000
	Solar PV System	25kW array + 100kWh battery + cabling	1	R1 000 000	R1 000 000
	TOTAL SYSTEM COST				R3 400 000